

Achieve Shorter Software Delivery Timelines Using Cloud IDEs

Using cloud services today has become a convenience. An integrated development environment (IDE) facilitates the development of apps. The authors of this article have checked out a number of cloud IDEs and evaluated them. Read on to discover which of these suit your requirements the best.

To develop a software product, the development process plays a vital role. Software development consists of requirement gathering, design, coding, testing, deployment and maintenance. Some of the basic requirements for software development are hardware and software tools, a server to deploy and test the application and a coding development environment. Cloud based development brings in a lot of cost savings and efficiency into the process. For example, the cloud provides a deployment environment, and so users do not require to build, develop or maintain this environment but can share it easily from the public environment. To leverage the customisation of software and hardware components, based on the deployment requirements, we need to move to the next generation cloud computing platform.

An integrated development environment (IDE) is a software tool that integrates various features required for the development of a software application. The latest improvement in this field is the cloud IDE, which is a virtual development environment. For example, to develop a Java application, we use Eclipse as the IDE, which provides an integrated plug-in facility to develop Java code, auto complete Java APIs, has syntax highlighting, apart from the capability to compile, execute, test and analyse static code. In a traditional software development environment, all developers install the IDE in their respective systems, and upload their developed code into a concurrent version system (CVS) or source

code management system like GIT. Once the code is checked in, it requires the consensus of other developers, which takes more time if the number of developers is high in a project. So, there is the fundamental need to provide a software development environment for a large number of developers to work together on a collaborative project, yet achieve a shorter delivery time. A cloud IDE is one of the best solutions to reduce the development process time, as it provides a more convenient and flexible common environment for software developers.

The cloud IDE

The key idea behind the cloud IDE based platform is to create a virtual development environment. It can provide the following features for the software development process:

- Users can develop an application in one or more software technologies such as Java, C++, HTML, JavaScript and so on.
- It has an application container to deploy the application on-the-fly.
- It offers repository management to store, retrieve and share code between groups of developers.

In addition to these basic features, we need to add security to limit the access to only the selected group of developers. This can be achieved by an authentication system that protects the development environment from public use.

Here, let's consider an open source virtual environment for cost effective and better community

